

Sanghyun Joshua Cho

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EDUCATION

Purdue University, College of Science

West Lafayette, IN

Bachelor of Science in Computer Science, Focus: Systems Engineering

May 2027

- **Relevant Coursework:** Object-Oriented Programming in Java, Discrete Mathematics, Programming in C, Computer Architecture, Data Structures and Algorithms, Advanced Algorithms, Compilers

EXPERIENCE

Electrical Engineering Systems Assistant | Python, Linux

October 2025 – Present

Purdue University

West Lafayette, IN

- Developing Python scripts to automate recurring facility data collection tasks from distributed sensor logs, transforming records into structured datasets that staff could directly utilize in planning with faculty.
- Automated routine reporting, boosting system availability by **12%** and cutting manual effort.
- Analyze operational datasets to identify usage trends and failure signatures across lab equipment, informing predictive maintenance strategies across multiple labs and directly enhancing the reliability of vital research systems.

Oathbound Studios | C++, Unreal Engine

May 2025 – October 2025

Systems Programmer

West Lafayette, IN

- Served as the primary point of contact for system design questions, providing guidance on best practices, object-oriented design patterns, and debugging strategies to a team of **20+** developers, helping elevate code quality.
- Built core gameplay systems in C++ using OOP and data-oriented design techniques, increasing modularity and maintainability while reducing heap allocation frequency and sharpening overall memory efficiency.
- Ported performance-critical Blueprint systems into native C++ classes, eliminating interpretation overhead and improving runtime loop latency by **16%** in targeted hot paths.

PROJECTS

Beatproof | Next.js, TypeScript, Solana

October 2025

- Led the Next.js + TypeScript architecture for Beatproof's Solana audio marketplace, pairing server-rendered React with Framer Motion animation and responsive UI engineering to improve first meaningful paint by **28%**.
- Architected Phantom wallet onboarding with Solana's proprietary wallet alongside secure Next.js session APIs, reducing wallet drop-offs and tightening successful marketplace entries during internal QA.
- Drove deep stack integration, optimizing Solana RPC calls with Anchor utilities, unifying Zod validation, and modernizing bundling/linting, which shortened regression test cycle and stabilized QA environments.

Heater App – iOS | Swift, Firebase

Aug 2025 – Oct 2025

- Built a SwiftUI iOS app for sports game logging using MVVM architecture, ensuring predictable UI state and maintainable code.
- Integrated Firebase Auth and Firestore to enable secure user authentication, real-time data persistence, and seamless cross-device synchronization, ensuring that user game logs and metadata were consistently reflected across devices.
- Deployed to a college community of over **750 users**, with a local server API for sports metadata (teams, schedules, stats) and caching that reduced redundant network requests by **21%**, improving responsiveness and load times.

Mini C Compiler | C/C++, Lex, Yacc

Sep 2025 – Present

- Developing a compiler in C++ for a simplified C-like language, emitting machine instructions on a Unix target.
- Implemented semantic analysis features that detect programming errors, track dataflow, and identify dead computations, enabling earlier elimination of unnecessary instructions and reducing code paths by **30%**.
- Utilized Lex and Yacc to implement the compiler's lexical and syntactic analysis stages, producing typed ASTs that feed into IR generation and allow for deterministic code translation and future optimization passes.

Unix Shell | C/C++, Lex, Yacc

Jul 2025

- Built a Unix-like command line in C and C++ on Ubuntu, leveraging Lex and Yacc for tokenization and parsing.
- Extended the shell with environment variable + wildcard expansion, background processes, fork/exec pipelines, proper signal handling, and subshell execution semantics, ensuring correct behavior across multi-process command chains.

SKILLS

Languages: C++, C, Swift, Python, Bash, Java, JavaScript

Technologies: Unreal Engine, Lex/Yacc, LLVM, Unix/Linux Systems, Next.js, Firebase, Phantom

DevOps / Tooling: Docker, Linux shell scripting, CMake, Make, GitHub Actions, Vercel

Concepts: Compiler design, Parsing, IR construction, POSIX processes and signals, Memory management, Memory allocation, Concurrency